Neural Networks and Deep Learning Aksel Kretsinger-Walters, adk2164
Lecture 13B Summary: Interpretability in Deep Learning
COMS 4776: Fall 2025 **Due:** — 11:59 PM ET

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1 Overview

Deep neural networks achieve impressive performance, yet their internal mechanisms remain difficult to interpret. This lecture explores methods for understanding model behavior through visualization, gradients, and analysis of learned representations.

2 Input Gradients

The gradient $\partial L/\partial x$ reveals the model's sensitivity to small input changes and can serve as a tool for interpretability or debugging. However, raw gradients are often noisy and emphasize local perturbations rather than meaningful global structure.

2.1 SmoothGrad

To reduce noise, SmoothGrad (Smilkov et al.) averages gradients across noisy perturbations of the same image:

$$S = \frac{1}{N}\sum_{i=1}^{N}\frac{\partial L}{\partial x}(x+\epsilon_i), \quad \epsilon_i \sim \mathcal{N}(0,\sigma^2 I)$$

This produces smoother, more globally interpretable saliency maps.

3 Feature Visualization

Visualization techniques attempt to understand what hidden units represent:

- **Activation maximization:** find input images that strongly activate a given neuron or layer.

- **Layer-wise feature maps:** earlier layers capture edges and textures; deeper layers respond to higher-level concepts.

Gradient ascent on images can be used to iteratively modify inputs to amplify certain activations, as demonstrated in the *Distill.pub* visualizations.

4 Adversarial Examples

Small, imperceptible perturbations can cause large changes in predictions. Given an image x labeled "cat," we can compute:

$$x' = x + \epsilon \cdot \mathrm{sign}\left(\frac{\partial L}{\partial x}\right)$$

to make the model misclassify it (e.g., as "dog"). This *Fast Gradient Sign Method (FGSM)* reveals that networks rely on fragile, non-human features.

Adversarial examples are transferable across models and persist even in physical settings (e.g., printed images or modified road signs), posing serious security challenges.

5 Deep Dream

The *Deep Dream* algorithm accentuates patterns already detected by a model by reinforcing activations during backpropagation:

$$x \leftarrow x + \eta \frac{\partial h_\ell}{\partial x}$$

where h_ℓ denotes activations at layer ℓ. Repeated updates make features "grow" in the image, producing surreal, highly textured visualizations.

6 Multimodal Interpretability

CLIP (Contrastive Language–Image Pretraining) aligns text and vision embeddings, enabling text-guided exploration of visual concepts and revealing semantically meaningful activations across modalities.

7 Sparse Representations

Superposition: concepts are distributed across multiple neurons. **Sparse coding hypothesis:** each concept is encoded by strong activation of a small subset of neurons. Two notions:

- **Population sparsity:** only a few neurons fire for each concept.
- **Lifetime sparsity:** each neuron activates for few inputs.

7.1 Sparse Autoencoders

Adding an ℓ_1 penalty or top-k sparsity constraint promotes more interpretable latent features. Recent work (e.g., *Transformer Circuits*) applies sparse autoencoders to transformer activations, discovering "monosemantic" neurons tied to human-interpretable concepts.

8 Circuits and Compositional Analysis

Interpretability extends beyond individual neurons to multi-layer **circuits**—structured pathways representing invariances or hierarchical features. For instance, neurons detecting "windows" and "wheels" combine to form a higher-level "car" detector, which then contributes to even more abstract semantic units.

9 Key Takeaways

- Input gradients and visualization offer insight but require careful interpretation.
- Adversarial examples expose model brittleness and highlight the gap between human and machine perception.
- Sparse and compositional representations hint that deep networks organize knowledge hierarchically and efficiently.